

MODULE 4.2

AND, OR, NOT Gates and Truth Tables

MODULE 4.2

Basic Logic Gates

AND Gate

OR Gate

NOT Gate

Truth Tables

Boolean Expressions

Gate Symbols

The AND Gate

ALL

Conditions must be true.

Example:

Key A **AND** Key B → Door opens



Real-Life AND Logic

Computer laboratory door security requirement:

 **Door Key**

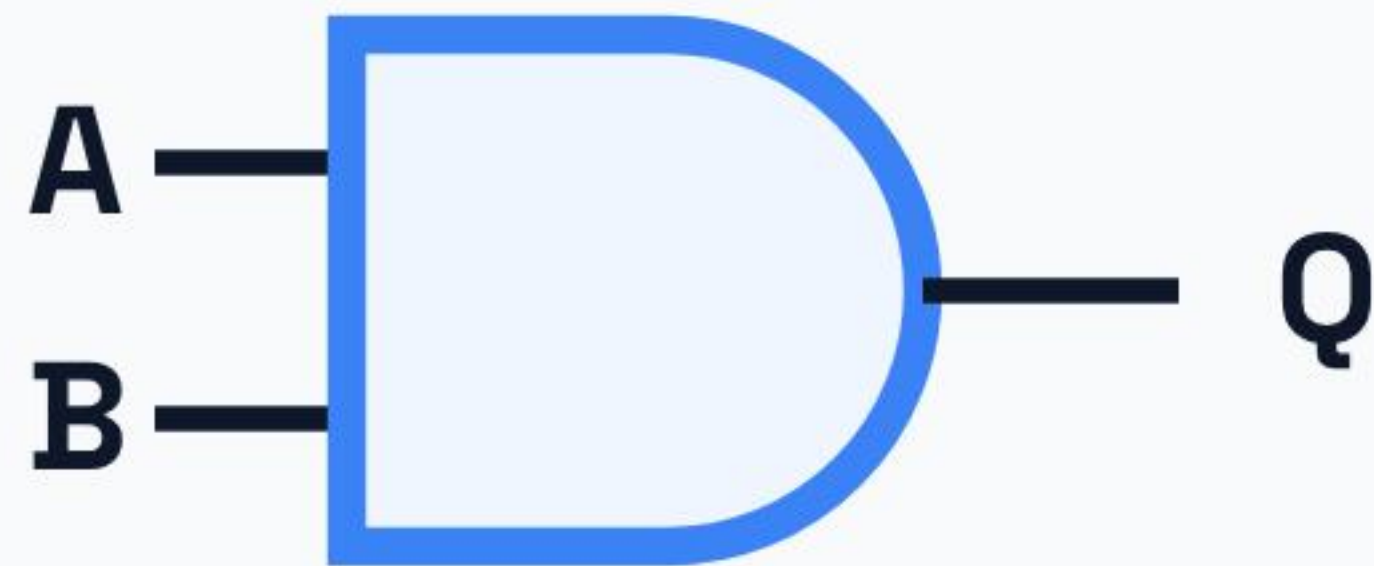
+ **AND**

 **Padlock Key**

If one key is missing → Door stays closed.

AND Gate Representation

Circuit Symbol



Boolean Expression

$$Q = A \cdot B$$

The dot (·) means AND

AND Gate Truth Table

A	B	$Q = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Output is 1 **only** when **ALL** inputs are 1.

MEMORY TRICK

AND = ALL

Are ALL inputs 1?

YES → 1

NO → 0

3-Input AND Gate

Boolean Expression

$$Q = A \cdot B \cdot C$$

Combinations

8 Possible
Rows (2^3)

The OR Gate

ANY

Condition can be true.

Example:

Front Door **OR** Back Door → Exit

Real-Life OR Logic

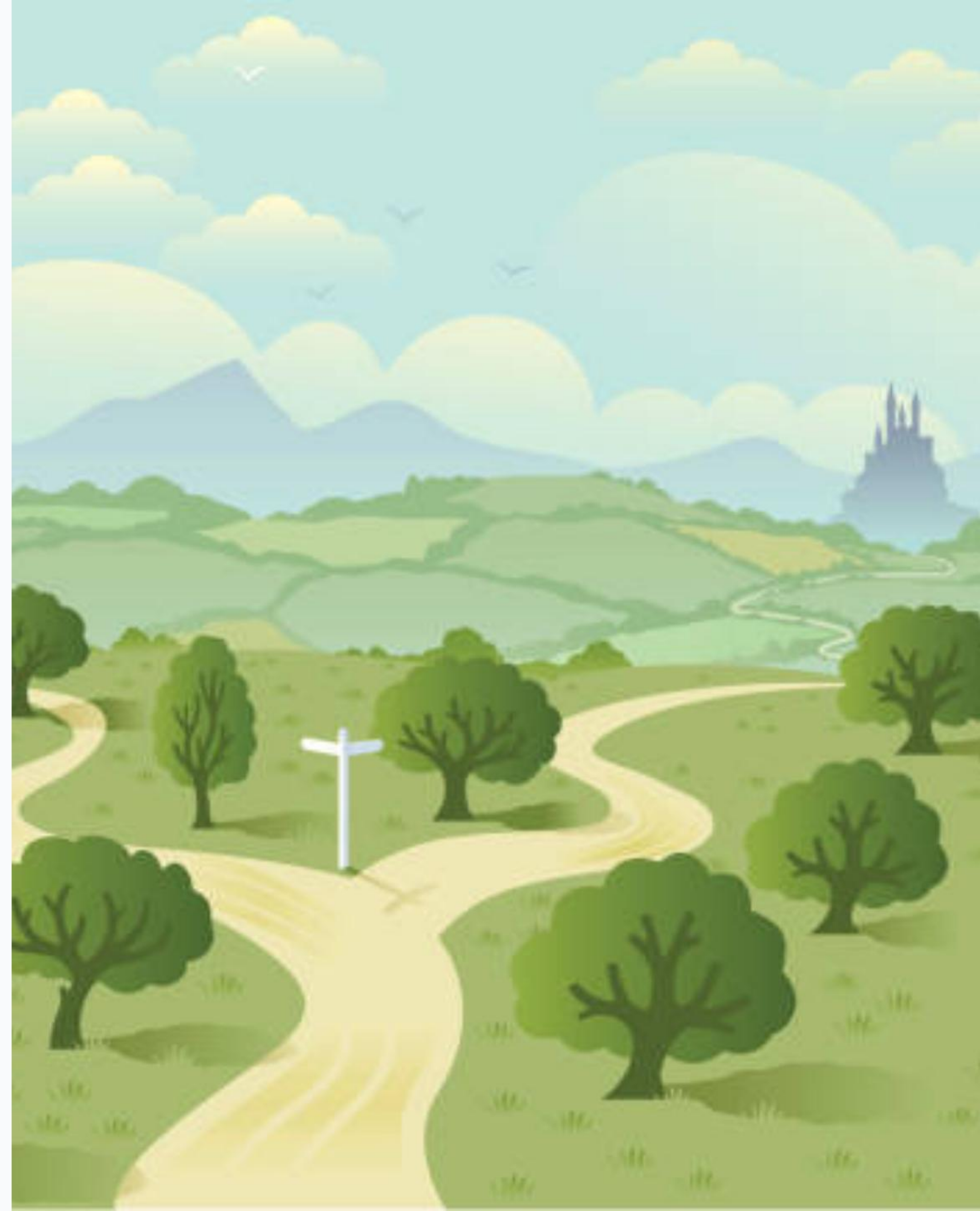
Reaching home via multiple routes:

 **Route A**

OR

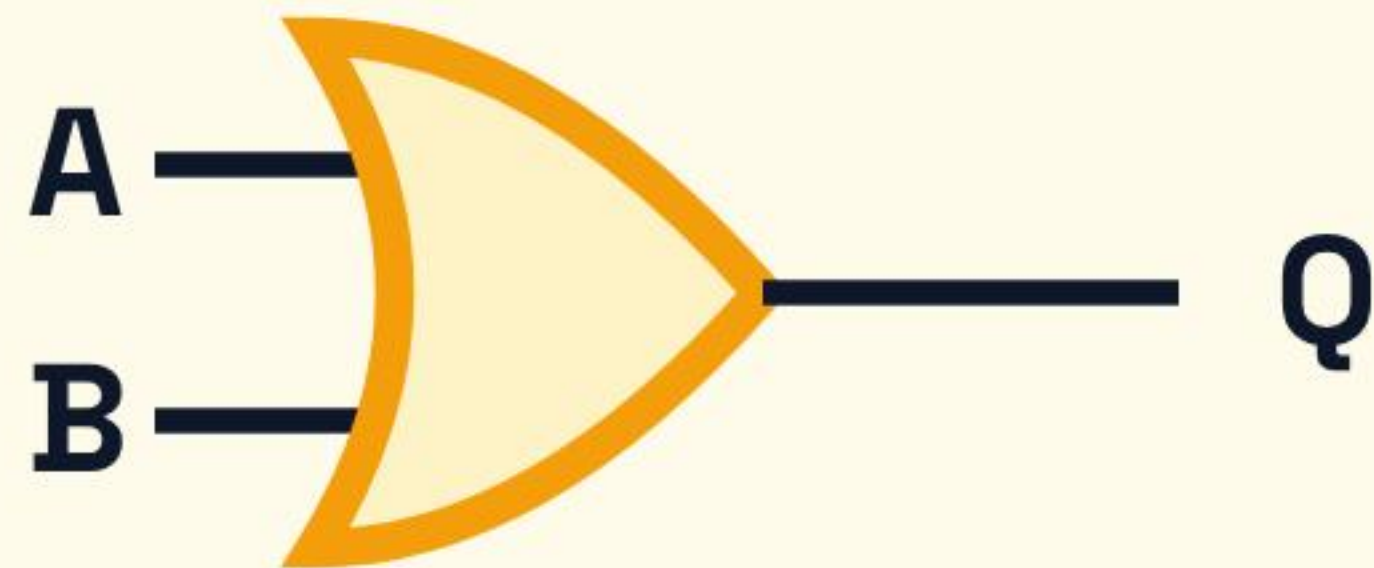
 **Route B**

Using either route succeeds.



OR Gate Representation

Circuit Symbol



Boolean Expression

$$Q = A + B$$

The plus (+) means OR

OR Gate Truth Table

A	B	Q = A + B
0	0	0
0	1	1
1	0	1
1	1	1

Output is 1 when AT LEAST ONE input is 1.

MEMORY TRICK

OR = ANY

Is at least ONE input 1?

YES → 1

NO → 0

The NOT Gate

OPPOSITE

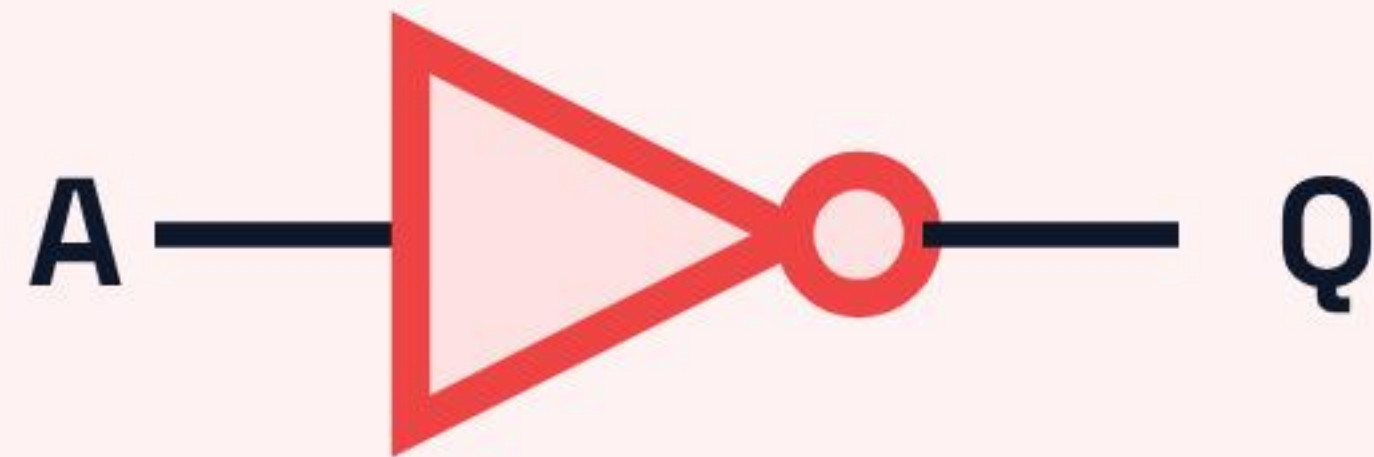
Produces the complement.

Input 0 → Output 1

Input 1 → Output 0

NOT Gate Representation

Circuit Symbol



Boolean Expression

$$Q = \bar{A}$$

The bar means NOT

NOT Gate Truth Table

A	$Q = \bar{A}$
0	1
1	0

NOT changes **0** → **1**

NOT changes **1** → **0**

Boolean Expression Summary

Gate	Expression	Main Rule
AND	$Q = A \cdot B$	ALL inputs must be 1
OR	$Q = A + B$	AT LEAST ONE input must be 1
NOT	$Q = \bar{A}$	Output is OPPOSITE

Gate Comparison

AND

INPUTS

2 or more

OUTPUT = 1 WHEN...

ALL inputs are 1

OR

INPUTS

2 or more

OUTPUT = 1 WHEN...

AT LEAST ONE is 1

NOT

INPUTS

Strictly 1

OUTPUT = 1 WHEN...

Input is 0

Understanding Truth Tables

A truth table shows all possible input combinations and their corresponding outputs.

2 Inputs

$$2^2 = 4 \text{ combinations}$$

3 Inputs

$$2^3 = 8 \text{ combinations}$$

Truth Table Practice

QUESTION:

For an AND gate, what is the output when **A = 1** and **B = 0**?

Answer: Q = 0

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Knowledge Check

1

Which gate gives 1 only when all inputs are 1?

A. OR

B. AND

C. NOT

2

Which gate gives the opposite output?

A. AND

B. OR

C. NOT

3

Output of OR when A=0 and B=1?

A. 0

B. 1